Keyword Vectorization for Retrieval Augmented Generation on Science Question Answering

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Abstract: This paper investigates novel retrieval techniques in closed domain question answering (QA) systems, aiming to enhance information retrieval and comprehension tasks. It looks into the use of standardised academic tests as benchmarks for evaluating Language Models (LLMs), shedding light on their proficiency levels in science-related subjects. By integrating these tests as evaluation metrics, efforts are made to augment LLM capabilities across diverse domains, including science education and research. Hence, this study introduces vectorised keyword search techniques for closed-domain science question answering, focusing on specialised keyword lists and three vectorisation methods to assess question retrieval efficacy.

1.0 Introduction

Language Models (LLMs) have emerged as powerful tools in natural language processing, demonstrating remarkable capabilities in understanding and generating human-like text. Particularly, their application in question answering (QA) systems has garnered significant attention due to their potential to automate information retrieval and comprehension tasks. LLMs have been employed in both open and closed domain QA scenarios. In open domain QA, LLMs are tasked with answering questions across a broad range of topics, often requiring comprehensive knowledge and reasoning abilities. Conversely, closed domain QA focuses on specific domains or datasets, facilitating more targeted information retrieval and analysis. This paper explores novel retrieval techniques in closed domain QA, examining their strengths, limitations, and potential applications in advancing information retrieval and comprehension tasks.

1.1 Academic Question Answering

The use of standardised academic tests like the SATs and AP exams as evaluation benchmarks for Language Models (LLMs) is a notable approach in natural language processing research. These assessments provide a structured framework for assessing LLM proficiency in comprehension and analysis tasks. Observing LLMs scoring around the 60th percentile in science-related subjects on these tests (Openai, 2024) highlights both advancements and remaining challenges in LLM development. While LLMs demonstrate competency in understanding scientific concepts, further refinement is necessary to enhance their proficiency levels. Integrating SATs and AP exams as evaluation metrics underscores efforts to augment LLM capabilities across diverse domains, including science education and research.

2.0 Related Work

Recent academic literature explores prompt engineering, focusing on refining prompts to optimise language model (LLM) performance. Strategies such as zero-shot and few-shot prompting are investigated to enhance LLM capabilities. Additionally, Retrieval Augmented Generation (RAG) emerges as a significant advancement, bridging retrieval-based and generative models to improve response quality and relevance across various domains.

2.1 Prompt Engineering

In this paper, we focus on two methods within the domain of prompt engineering: few-shot prompting and Chain of Thought (CoT) prompting. Few-shot prompting is a methodology employed in machine learning, where a model is trained with a limited number of examples specific to a given task. This approach involves supplying the model with a minimal set of input-output pairs, enabling it to generalise and proficiently perform similar tasks. A notable study has demonstrated a notable enhancement in answer accuracy, showcasing a remarkable 35% increase with just one-shot prompting, and up to a substantial 50% increase with a ten-shot prompt (Brown et al., 2020). Another technique, known as Chain of Thought (CoT) prompting, involves guiding a language model through a series of logical steps mirroring human reasoning patterns. Through the provision of sequential examples illustrating a logical progression, the model acquires the ability to establish connections between concepts and produce coherent responses (Wei et al., 2023).

2.2 Retrieval Augmented Generation

Retrieval Augmented Generation (RAG) represents a substantial advancement in the integration of retrieval-based and generative natural language processing (NLP) models (Lewis et al., 2021). Fundamentally, RAG functions by parsing input texts, segmenting them into discrete units, and embedding them for streamlined retrieval. During runtime operation, user queries are embedded, top-k relevant results are retrieved, and suitable text segments are selected based on predefined retrieval strategies. This hybrid approach not only enhances the efficiency of information retrieval but also facilitates the generation of coherent responses. Thus, RAG stands as a promising methodology in bridging the gap between retrieval-based and generative NLP paradigms.

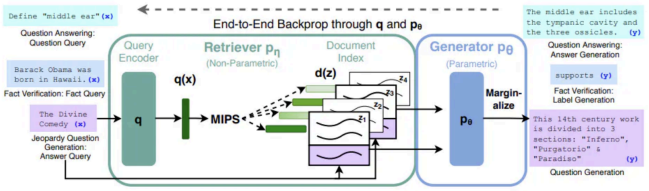


Diagram 1: Retrieval Augmented Generation
(Lewis et al., 2021)

Key to RAG's functionality is its hierarchical index retrieval system, which enables efficient storage and retrieval of text chunks. Additionally, RAG employs advanced retrieval techniques such as context enrichment through sentence window retrievers and parent document retrievers, as well as fusion retrieval methods combining keyword search and vector search. Furthermore, RAG incorporates query transformations, query routing, and multi-document agents to enhance response quality and relevance. The system is underpinned by fine-tuning mechanisms for encoder, ranker, and LLM components, ensuring optimal performance across various applications. Moreover, RAG finds applications in diverse domains, from knowledge-intensive multistep questions to instruction tuning and in-context learning. As the field of RAG continues to evolve, it promises to revolutionise natural language

understanding and generation tasks, paving the way for more sophisticated and contextually relevant AI applications.

2.3 Keyphrase Generation

Keyphrase generation, a critical task in natural language processing (NLP), is characterised by several distinct methodologies. (Çano & Bojar, 2019) Extractive Keyphrase Generation involves the application of supervised methods, such as classifiers leveraging document terms like IDF, or employing graphical models like PageRank. Keyphrase Datasets, essential for model training, are typically constructed using dense models trained on publicly available datasets, such as StackExchange for post topics or TextWorld for game observations and commands. Abstractive Keyphrase Generation, another prominent approach, utilises sophisticated architectures like encoder-decoder models such as CopyRNN, or hybrid solutions integrating encoder-decoder LSTM models to enhance the accuracy of keyphrase extraction.

3.0 Problem Formulation

While LLMs show promise in addressing complex science questions, their effectiveness is hindered when confronted with queries involving tables, graphs, and intricate data structures. Existing approaches, such as using few-shot Chain of Thought prompting by storing examples in either a vector store or keyword reference store, have limitations due to the complexity of the questions and the noise created by the sheer length of the questions with long contextual passages. For instance, the vector store approach might overlook nuanced semantic relationships crucial for understanding scientific concepts, while the keyword reference store approach may lack the adaptability to capture the diverse attributes of scientific inquiries. As a result, there is a need to explore alternative techniques to enhance the retrieval and comprehension of science-related questions by LLMs.

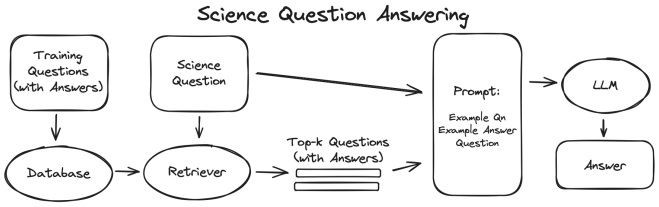


Diagram 2: Science Question Answering with a RAG

In this specific scenario, science question answering is conducted utilising LLMs, employing a one or two-shot Chain of Thought prompting approach. The training questions are incorporated into the database and subsequently retrieved based on a new science question from the evaluation question list. The top 1 or top 2 ranked retrieved question is then integrated into the prompt preceding the actual question, formatted as "Example Question, Example Answer, Question," with the expectation that the LLM will continue by answering the latter question. The objective is to not only explore existing retrieval methods but also to experiment with a novel approach aimed at enhancing the effectiveness of few-shot CoT prompting through the retrieval of superior, more relevant examples.

4.0 Methodology

In this study, vector search and keyword vector search methodologies are introduced as techniques to enhance question retrieval efficiency and comprehension. This is then contrasted with the novel approach of a vectorised keyword search for closed-domain science question answering.

4.1 Data Selection

Among the array of publicly available science QA assessments, a test was selected based on specific criteria. The criteria for selecting the test encompass three key points. Firstly, the chosen test should highlight areas where current cutting-edge LLMs demonstrate inadequacies, despite efforts in prompt engineering and document retrieval systems. Secondly, preference is given to a test that can be substantially enhanced through improvements in document retrieval methodologies, rather than relying primarily on other modal transformations such as image-to-text conversions. Lastly, the test should ideally demand minimal data cleaning and processing, thereby streamlining the preparatory efforts required prior to conducting analyses.

Based on the aforementioned criteria, we have opted for the Science section of the ACT test (“ACT Test Prep,” n.d.), widely utilised as a university entrance examination. These assessments emphasise logical reasoning and deduction, drawing upon candidates' existing scientific knowledge, including comprehension of chemical equations, and the ability to deduce from presented scientific scenarios, often involving measurements and experiments. These tests can therefore be improved by both the provision of relevant knowledge in the prompt, as well as Chain of Thought prompting.

4.2 Data Processing

The following data types are collected from the ACT questions and answers, with detailed information provided below. Most questions contain a Context, encompassing details such as experiment depictions. This segment involves several processing steps, including transforming bar graphs into TeX-formatted tables, diagrams into descriptions, and equations into TeX-formatted equations. All context data exclusively support text data and TeX formats, selected for its compatibility with ChatGPT LLMs. Questions are also formatted in TeX if necessary and linked to their context in a one-to-many relationship (one question to many contexts). The Answer consists of a single letter indicating the MCQ choice selection, facilitating easy verification of correctness. The Explanation comprises the answer given with comprehensive logical justifications provided by PrepSharp (“ACT® Answer Keys,” n.d.), serving as prompts for Chain of Thought.

Data Type	Total	Details
Context	411	Graphs to TeX formatted tables, diagrams to diagram descriptions, equations to TeX formatted equations. All in text or TeX formatting.
Question	522	Text based question with TeX based formula processing, and multiple choice selection. Every Question has 1 Answer, 1 Explanation, and one-to-many Context.
Answer	522	Single capitalised alphabet for the MCQ answer
Explanation	522	Detailed explanation in text for Chain of Thought prompting

Table 1: Data Types generated after Cleaning

4.3 Vector Search

Vector Search serves as a fundamental strategy for Retrieval Augmented Generation, where we implement a Parent Document Retriever approach (LangChain, n.d.), dividing documents into child chunks. In splitting documents for retrieval, we face conflicting needs: smaller documents for accurate embeddings and longer documents to maintain context. The Parent Document Retriever achieves this balance by segmenting and storing small data chunks, initially retrieving them during retrieval and then obtaining the associated parent IDs to retrieve the corresponding larger documents.

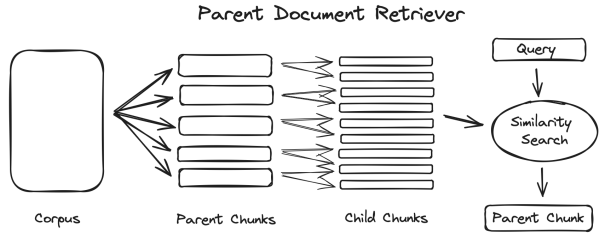


Diagram 3: Parent Document Retriever

This approach follows a sequential process: (1) Chunking into parent and child chunks, (2) Embedding the child chunks referencing the parent chunks, (3) Retrieving parent chunks via vector similarity assessment between child chunks and query, and (4) Utilising the top-n parent chunks as contextual information for the prompt. This method of retrieval is employed to identify the most similar question, facilitating Chain of Thought prompting. The step-by-step Answer and Explanation provided serve as a few-shot illustration of how to address the question effectively.

4.4 Keyword Vector Search

The average word count of a question is 361 words, constituting the aforementioned parent chunk. While this sizable unit can potentially be subdivided into smaller segments, such an approach risks diluting the holistic context of the question. However, this challenge can be effectively addressed through the utilisation of the Keyword Vector Search, also recognized as the Hybrid Search technique (LangChain, n.d.-a), which entails leveraging both sparse retrievers, like BM25, and dense retrievers, such as embedding similarity, to maximise retrieval accuracy. By integrating these diverse retrieval methodologies, the Keyword Vector Search enhances the system's ability to capture nuanced semantic relationships and contextual nuances, thereby yielding more precise and comprehensive search results.

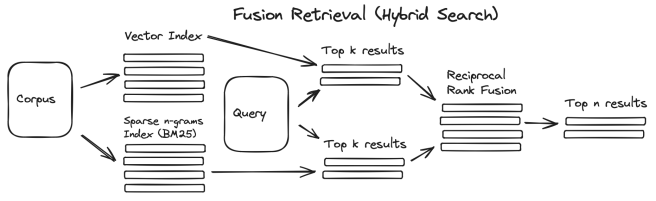


Diagram 4: Hybrid Search

This amalgamation of sparse and dense retrievers is particularly crucial in the context of science question answering with few-shot retrieval. In scientific domains, questions often require nuanced understanding and contextual reasoning, making it essential to retrieve relevant information accurately and efficiently. Sparse retrievers, such as BM25 (Robertson, Zaragoza, & Taylor, 2004), excel at capturing keyword-based relevance, enabling the identification of documents based on specific terms or phrases. However, they may overlook semantic similarities or fail to grasp the broader context of the query.

On the other hand, dense retrievers, like embedding similarity, are adept at understanding semantic relationships and capturing contextual nuances. They can identify documents based on similarities in meaning, even when the exact terminology varies. By combining the strengths of both sparse and dense retrievers, the Keyword Vector Search technique enhances the system's capability to address a wide range of queries effectively, facilitating more accurate and insightful answers to scientific questions with limited training data.

$$RRFscore(d \in D) = \sum_{r \in R} \frac{1}{k + r(d)}$$

To further optimise the retrieval process and ensure the most relevant documents are prioritised, the Keyword Vector Search employs the Reciprocal Rank Fusion (RRF) algorithm (Cormack, Clarke, & Büttcher, 2009). RRF combines the rankings produced by multiple retrieval systems by assigning weights based on the reciprocal ranks of documents. Documents that are ranked higher across multiple retrieval systems are given higher weights, leading to a final ranking that maximises the relevance of the retrieved documents. This integration of RRF enhances the effectiveness of the Keyword Vector Search by synthesising the strengths of diverse retrieval strategies and improving the overall accuracy of the retrieval process. In this specific experiment, the constant k is set to 1 to mitigate the disproportionate influence of higher-ranked weights.

4.5 Vectorised Keyword Search

In contrast to conventional keyword-based retrievers relying on the corpus's keywords, our approach for the closed-domain science question answering task involves crafting specialised keyword lists. These lists aim to encapsulate the diverse attributes of questions through meticulous keyword selection. Our devised keyword lists prioritise the following criteria: (1) commonly performed experiments and measurement techniques, (2) the specific science topic addressed in the question, and (3) keywords recurrently found in past papers.

Keyword List	Total	Examples
Science Topics	50	Enzyme kinetics, Newton's laws, Atomic structure, Chemical reactions, Kinetic energy, Electromagnetic spectrum, Genetics
Experiments	14	Interpretive analysis, Statistical inference, Observational analysis, Tabular scrutiny
Common Keywords	468	Average, experiment, mass, figure, results, concentration, \hline, study, g, temperature

Table 2: Keyword Lists

For each question, a set of keywords is chosen from the keyword lists to represent the question in a vector store. Alongside selecting keywords from the list, an LLM-generated summary of the question is also incorporated as a final addition to the keyword representation.

This list of keywords serves as the basis for evaluating the effectiveness of three distinct vectorization methods in question retrieval. Priority is given to identifying the most similar question

with a coherent logical progression to achieve optimal results. The vectorization techniques encompass: (1) amalgamating all keywords into a unified string, (2) amalgamating the top-k keywords from each keyword list individually before consolidating them into a singular vector store, and (3) amalgamating the top-k keywords from each keyword list individually before storing them in separate vector stores for individual retrieval. The RRF algorithm is then utilised to combine their distinct rankings and determine the highest value.

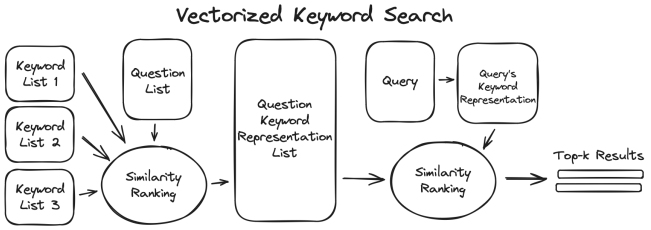


Diagram 5:Vectorised Keyword Search

The procedure commences with the generation of keywords for each question. This is accomplished through a similarity ranking of keywords in the keyword lists and the list of questions, which are then subjected to vectorization using three distinct keyword representation methods and stored in the vector store. Following this, when a new science question emerges, its keywords are generated, and a keyword similarity search is performed using the three respective methods to retrieve the top-k results. These outcomes serve as few-shot examples in the prompt to augment the quality of answers. The responses are subsequently refined and validated against an answer sheet for accuracy.

5.0 Evaluation

The experiment employed LLMs with gpt-3.5-turbo, using Faiss as the vector retriever and BM-25 as the keyword ranking algorithm. Results showed that the Vectorised Keyword Search, especially the Individual Keywords approach, outperformed other methods, emphasising the correlation between efficacy and optimal keyword matches. Human evaluation indicated a trade-off between question and answer similarity across different retrieval methods, suggesting the need for further refinement to improve overall performance.

5.1 Experimental Details

In this experiment, evaluation entails the utilisation of two papers comprising 76 questions, designated as evaluation queries, while the remaining 12 papers containing 446 questions function as the training queries stored in a retrieval database. The following delineates the constants and experimental specifics for the aforementioned experiment.

Firstly, the LLMs employing gpt-3.5-turbo are utilised. Subsequently, the embeddings utilised throughout the experiment are sourced from the OpenAI embeddings model text-embeddings-3-large. The vector retriever employed is Faiss (Facebook AI similarity search) (Facebook, n.d.), a similarity search algorithm facilitating the retrieval of up to the kth nearest neighbours with efficiency enhancements. The keyword ranking algorithm utilised is BM-25, a bag of words retrieval function incorporating idf (inverse document frequency) weights. The rank reciprocal fusion function employed here is the sum of 1 / (k + rank(i)), where k = 1 in this experiment. The detailed constants encompass a temperature of 1, a top-p of 0.7, and a top-k of 32 where applicable. A divergence occurs in the top-k results utilised in the vectorized keyword search for question keyword representation, which employs a k value of 3, signifying a three-keyword representation.

5.2 Case Study

In the preceding experiment, an analysis was conducted on a question that yielded an incorrect answer through Keyword Vector Search but a correct answer through Vectorised Keyword Search. This analysis aimed to elucidate the underlying processes and rationale behind the improvements implemented in this experiment. The ensuing excerpt presents the original question alongside one-shot examples provided by Keyword Vector Search and Vectorised Keyword Search, accompanied by their respective answers and explanations.

Question: According to the results of the study, did the injection of CCP grout reduce the \[SO_{4}^{2-} \] content of the groundwater flowing out of the mine?
F. Yes; \[SO_{4}^{2-} \] content was much lower on January 1,1996, than it was on July 1, 1995.
G. Yes; \[SO_{4}^{2-} \] content was much lower on January 1, 1997, than it was on July 1, 1996.
H. No; \[SO_{4}^{2-} \] content was much higher on January 1, 1996, than it was on July 1, 1995.
J. No; \[SO_{4}^{2-} \] content was much higher on January 1, 1997, than it was on July 1, 1996.

Diagram 6: ACT science Question 1

Question: According to the results of the study, the greatest difference between the sediment concentrations of the samples collected at the same time from the 2 flumes was closest to which of the following?
A. 2 g/L
B. 4 g/L
C. 6 g/L
D. 8 g/L

Diagram 7: ACT science Question 2

Question: When each of the 4 halide solutions is mixed with the \[Ca^{2+} \] test solution, only 1 precipitate forms. This precipitate most likely has what chemical formula?
F. CaF
G. \[Ca_2F \]
H. \[CaF_2 \]
J. \[CaF_3 \]

Diagram 8: ACT science Question 3

According to the results of the study, the injection of CCP grout did reduce the \(SO_{4}^{2-} \) content of the groundwater flowing out of the mine. We need to find the dates where there's a significant decrease in \(SO_{4}^{2-} \) content after the injection of CCP grout. Looking at the provided data:
- Before the injection (up to July 1, 1996), the \(SO_{4}^{2-} \) content remains relatively stable around 1140-1150 mg/L.
- After the injection (starting from January 1, 1997), there's a noticeable decrease in \(SO_{4}^{2-} \) content, dropping to 1060 mg/L on January 1, 1997, and continuing to decrease thereafter.
So, according to the results:
- F. Yes; \(SO_{4}^{2-} \) content was much lower on January 1, 1996, than it was on July 1, 1995. (This statement doesn't correspond to the data.)
- G. Yes; \(SO_{4}^{2-} \) content was much lower on January 1, 1997, than it was on July 1, 1996. (This statement corresponds to the data.)
- H. No; \(SO_{4}^{2-} \) content was much higher on January 1, 1996, than it was on July 1, 1995. (This statement doesn't correspond to the data.)
- J. No; \(SO_{4}^{2-} \) content was much higher on January 1, 1997, than it was on July 1, 1996. (This statement doesn't correspond to the data.)
Therefore, the correct answer is:
G. Yes; \(SO_{4}^{2-} \) content was much lower on January 1, 1997, than it was on July 1, 1996.

Diagram 9: ACT science Answer 1

To find the greatest difference between the sediment concentrations of the samples collected at the same time from the two flumes, we need to compare the sediment concentrations with and without the slurry layer at each time interval.
Let's calculate the differences in sediment concentrations at each time interval:
\[
\begin{align*}
\text{Time (min)} & : \text{Difference in Sediment Concentration (g/L)} \\
0 & : 0 - 0 = 0 \\
5 & : 4 - 0 = 4 \\
10 & : 4.5 - 3.5 = 1 \\
15 & : 5 - 3.5 = 1.5 \\
20 & : 4 - 5 = -1 \\
25 & : 3.5 - 7.5 = -4 \\
30 & : 3.5 - 9 = -5.5 \\
35 & : 3.5 - 10.5 = -7 \\
40 & : 3.5 - 10 = -6.5 \\
45 & : 3.5 - 10 = -6.5 \\
\end{align*}
\]
The greatest difference in sediment concentration occurs at \(t = 35 \) minutes, with a difference of \(3.5 - 10.5 = -7 \) g/L. However, since we are interested in the absolute difference, the answer would be \(|-7| = 7 \) g/L. So, among the options provided: D. 8 g/L Would be the closest to the greatest absolute difference in sediment concentrations.

Diagram 10: ACT science Answer 2

Answer: H
Explanation: 1. The prompt states that when each of the 4 halide solutions (F^-, Cl^-, Br^-, I^-) is mixed with the \[Ca^{2+} \] test solution, only 1 precipitate forms.
2. From Table 1, we can see that when F^- is mixed with \[Ca^{2+} \], a shiny white precipitate forms. This eliminates options G and J as they do not contain F.
3. When Cl^- is mixed with \[Ca^{2+} \], no precipitate forms. This eliminates option F as it contains F.
4. When Br^- is mixed with \[Ca^{2+} \], no precipitate forms. This eliminates option J as it contains F.
5. When I^- is mixed with \[Ca^{2+} \], no precipitate forms. This eliminates option J as it contains F.
6. From the above deductions, we can conclude that the precipitate formed when each of the 4 halide solutions is mixed with the \[Ca^{2+} \] test solution is most likely \[CaF_2 \], which is option H.

Diagram 11: ACT science Answer 3

In the Keyword Vector Search, Question 2 and its corresponding answer were retrieved as a one-shot example. The rationale behind this is evident in Question 1, which focuses on groundwater, specific dates, and tables and specific measurements in millimetres within the question's context. The vector database and keyword database identify the incorrect keywords and information, thereby relating the two questions. However, discrepancies arise when examining their corresponding answers. Answer 1 discusses the presence of sulphate ions and pH readings indicating their presence, while Answer 2 pertains to sediment concentrations, with the table providing readings of runoff and slurry levels.

In the Vectorised Keyword Search, particularly using the Individual Keywords representation, the keywords representing Question 1 are as follows (3 keywords from each of the 3 keyword lists): Atomic structure, Chemical reactions, Periodic table, Tabular scrutiny, Interpretive analysis, Data presentation, Concentration, Solution, Rate.

Conversely, Question 3 generated the following keywords: Chemical reactions, Periodic table, Thermodynamics, Tabular scrutiny, Data-driven insights, Data presentation, Acid, Mixture, Reaction.

The similarity in keywords and their selection becomes apparent. Answer 3 also focuses on chemical reactions and formulas, akin to Answer 1, and delves into the presence of calcium ion precipitates in water.

5.3 Results

The results are tabulated based on the number of correct answers produced by the LLM out of the 76 evaluation questions posed. Both one-shot and two-shot examples were attempted, as deteriorating results were observed due to prompt length; each question comprises approximately 2000 characters, and each answer about 1000 characters. On average, a one-shot prompt consists of about 5000 characters, and a two-shot prompt about 8000 characters.

Regarding the representation strategies of the Vectorised Keyword Search, the terminology has been abbreviated as follows: (1) All Keywords - the amalgamation of all keywords into a unified string, (2) Individual Keywords - the amalgamation of the top-k keywords from each keyword list individually before consolidating them into a singular vector store, and (3) RRF individual keywords - the amalgamation of the top-k keywords from each keyword list individually before storing them in separate vector stores for retrieval using the RRF algorithm.

Evaluation	Correct Answers (zero-shot / one-shot CoT)	Correct Answers (two-shot CoT)
Question Only	53 / 76 (69.7%)	-
Vector Search	48 / 76 (63.1%)	48 / 76 (63.1%)
Keyword Vector Search	58 / 76 (76.3%)	55 / 76 (72.3%)
Keyword Search	56 / 76 (73.6%)	55 / 76 (72.3%)
Vectorised Keyword Search (All Keywords)	51 / 76 (67.1%)	51 / 76 (67.1%)
Vectorised Keyword Search (Individual Keywords)	60 / 76 (78.9%)	54 / 76 (71.0%)
Vectorised Keyword Search (RRF Individual Keywords)	58 / 76 (76.3%)	48 / 76 (63.1%)

Table 3: No of correct answers for each QA implementation

Regarding the Human evaluation, the Question and Answer similarities for each of the 6 Searches presented below are ranked manually for the 76 evaluation questions. Each question's searches were ranked from 1 to 6, with 1 indicating the highest similarity and 6 representing the greatest disparity. The cumulative scores for all evaluations were then aggregated.

Evaluation	Question Similarity (Human Evaluation)	Answer Similarity (Human Evaluation)
Vector Search	252 (1st)	278 (5th)
Keyword Vector Search	259 (2nd)	239 (1st)
Keyword Search	264 (4th)	266 (3rd)
Vectorised Keyword Search (All Keywords)	292 (6th)	299 (6th)
Vectorised Keyword Search (Individual Keywords)	260 (3rd)	245 (2nd)
Vectorised Keyword Search (RRF Individual Keywords)	269 (5th)	269 (4th)

Table 4: Question and Answer Similarity (Human Evaluation)

5.4 Discussion

Firstly, employing prompts with only questions establishes a baseline of approximately 70% accurate answers generated. However, the naive vector search algorithm, namely the Parent Document Retriever, achieves an accuracy score of about 63%. This suggests that either the length of the prompt is too long, resulting in a poorer response, or that the few-shot retrieval provides an incorrect example in this instance, biasing the answer in the wrong direction.

Next, we observe drastically worse results with the two-shot approach. As mentioned earlier, the one-shot prompt averages 5000 characters, while the two-shot prompt averages 8000 characters. Moreover, through visual analysis, considering that the training data comprises only 12 papers, each of which generally avoids repetition of questions on similar topics, the disparity in similarity scores between the top and second-top questions typically ranges from 0.05 to 0.1, indicating a notably broad distribution. Therefore,

for this discussion, we will focus more on the one-shot prompts which have proven to yield better results.

The Keyword Vector Search exhibits better performance, ranking highest among the pre existing approaches. Additionally, the lower Keyword Search score indicates the critical importance of integrating keyword and vector search methods for the algorithm's success.

In the evaluation of search methodologies, the Vectorised Keyword Search reveals distinct performance trends. Notably, the Individual Keywords approach emerges as the most effective, followed by the RRF Individual Keywords approach, with the All Keywords approach exhibiting the lowest performance. This hierarchy suggests that the efficacy of the algorithm correlates positively with its ability to select the optimal match between any two keyword groups. However, the comparatively lower performance of the RRF Individual Keywords method indicates potential challenges arising from suboptimal keyword selection strategies. Furthermore, it is conceivable that the underperformance of the cumulative results is influenced by the presence of a single keyword list exerting a disproportionately negative impact. Improved keyword selection techniques are therefore anticipated to yield more favourable outcomes.

The human evaluation of question versus answer similarity values reveals distinct trends across different methods. Specifically, methodologies employing vector search consistently yield higher scores for question similarity but lower scores for answer similarity. Conversely, the Vectorised Keywords method demonstrates higher answer similarity and lower question similarity. This observation suggests that the Vectorised Keywords method may prioritise achieving higher answer similarity during its retrieval process. However, it is noteworthy that while the rankings suggest a clear distinction between methods, the actual differences in answer similarity values are less pronounced. Many answer similarity values remain unchanged even with the implementation of the Vectorised Keyword Search technique, indicating potential limitations or complexities in the retrieval process that warrant further investigation.

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